

Q-3	(A)	Java code Filename: SmartHub.java	[09]
	1	<pre> import java.util.Scanner; interface Device { void turnOn(); void turnOff(); boolean getStatus(); } abstract class HomeAutomation { abstract void addDevice(Device device); abstract void removeDevice(Device device); abstract void checkDeviceStatus(Device device); abstract void turnOffAllDevices(); } class Light implements Device { boolean status; public void turnOn() { System.out.println("Turning the light ON..."); status = true; } public void turnOff() { System.out.println("Turning the light OFF..."); status = false; } public boolean getStatus() { return status; } } class SecurityCamera implements Device { boolean status; public void turnOn() { System.out.println("Activated security camera"); status = true; } public void turnOff() { System.out.println("Deactivated security camera"); status = false; } public boolean getStatus() { return status; } } </pre>	

```

class DeviceNotFoundException extends RuntimeException {
    DeviceNotFoundException(String msg) {
        super(msg);
    }
}

class InvalidChoiceException extends Exception {
    InvalidChoiceException(String msg) {
        super(msg);
    }
}

class SmartHub extends HomeAutomation {
    static Device l = new Light();
    static Device s = new SecurityCamera();

    public void addDevice(Device device) {
        if (device instanceof Light)
            System.out.println("Light added");
        else if (device instanceof SecurityCamera)
            System.out.println("Security camera added");
        else
            System.out.println("Something wrong in argument");
    }

    public void removeDevice(Device device) {
        if (device instanceof Light)
            System.out.println("Light removed");
        else if (device instanceof SecurityCamera)
            System.out.println("Security camera removed");
        else
            System.out.println("Something wrong in argument");
    }

    public void checkDeviceStatus(Device device) {
        System.out.println(device.getStatus());
    }

    public void turnOffAllDevices() {
        System.out.println("turning off all devices");
        l.turnOff();
        s.turnOff();
    }

    static void handleUserChoice(int choice) throws InvalidChoiceException {
        throw new InvalidChoiceException("invalid choice");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            int choice;
            boolean lightAdded = false;
            boolean cameraAdded = false;

```

```

do {
    System.out.println("Enter choice from menu:");
    System.out.println("1. Add light");
    System.out.println("2. Add security camera");
    System.out.println("3. Turn lights on/off");
    System.out.println("4. Turn security camera activate/deactivate");
    System.out.println("5. check status of light");
    System.out.println("6. check status of security camera");
    System.out.println("7. turn of all device");
    System.out.println("8. Exit");
    choice = sc.nextInt();

    switch (choice) {
    case 1:
        System.out.println("Adding light...");

        HomeAutomation hl = new SmartHub();
        hl.addDevice(1);
        lightAdded = true;
        break;

    case 2:
        System.out.println("Adding camera...");

        HomeAutomation hs = new SmartHub();
        hs.addDevice(s);
        cameraAdded = true;
        break;

    case 3:

        if (l.getStatus()) {
            l.turnOff();
        } else if (!l.getStatus()) {
            l.turnOn();
        }
        break;

    case 4:

        if (s.getStatus()) {
            s.turnOff();
        } else if (!s.getStatus()) {
            s.turnOn();
        }
        break;

    case 5:
        System.out.println("checking light status");
        if (!lightAdded) {
            throw new DeviceNotFoundException("No lights added");
        } else {
            HomeAutomation hcheck = new SmartHub();
            hcheck.checkDeviceStatus(1);
        }
    }
}

```

```

        break;

    case 6:
        try {
            System.out.println("checking camera status");
            if (!cameraAdded) {
                throw new DeviceNotFoundException("No cameras
added");
            } else {
                HomeAutomation hcheck = new SmarthHub();
                hcheck.checkDeviceStatus(s);
            }

        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
        break;

    case 7:
        HomeAutomation hoff = new SmarthHub();
        hoff.turnOffAllDevices();
        break;

    case 8:
        System.out.println("Exiting..");
        break;

    default:
        handleUserChoice(choice);
    }
} while (choice != 8);

} catch (Exception e) {
    System.out.println(e.getMessage());
} finally {
    System.out.println("Thank you for choosing SmarthHub");
}
}
}

```

ALL THE BEST